BERZIET UNIVERSITY
FACULTY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER ENGINEERING

ENCS5341

MACHINE LEARNING AND DATA SCIENCE

Assignment#3:

Prepared by:

Name: Jouwana Daibes	Number: 1210123	Section: 1
Name: Emil Khoury	Number: 1211223	Section: 2

Instructor:

Dr. Yazan Abu Farha

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Introduction

This report addresses a multi-task classification problem, where the objective is to simultaneously predict two categorical outputs: Activity and Time of Day, based on image data. The Activity task involves predicting various activities such as sightseeing, relaxing, and hiking, while the Time-of-Day task involves predicting whether the activity occurs in the morning, afternoon, or evening. These two tasks were chosen due to their complementary nature, providing valuable insights into the dataset and enhancing the overall prediction model.

To tackle this problem, we explore three machine learning models: k-Nearest Neighbors (k-NN), Random Forest (RF), and Convolutional Neural Networks (CNN). The k-NN model serves as a baseline, while the Random Forest and CNN models incorporate more advanced techniques for handling the complexity of the tasks. The report also highlights challenges such as class imbalance, particularly in the Activity classification task. The primary goal is to evaluate and compare the performance of these models in predicting Time of Day and Activity, and to explore methods for improving classification accuracy.

Exploratory Data Analysis

We preprocessed the dataset by cleaning and normalizing categorical labels, ensuring image labels matched by downloading them in order from the .csv file, and handling missing or failed downloads. Non-standard Time-of-Day entries were removed, missing activity labels were replaced with "No Activity," and all categorical fields were standardized to lowercase. These steps minimized noise for reliable analysis. We then performed exploratory data analysis (EDA) and statistics as shown in *Table1*. A Cramér's V heatmap is used as shown in *Figure1* below to examine relationships between categorical variables. The heatmap revealed weak to moderate associations, indicating complementary information, which supports multi-task learning. Visualizations like bar charts and stacked bar charts shown in *Figure2* further explored category distributions and associations between activity and time of day.

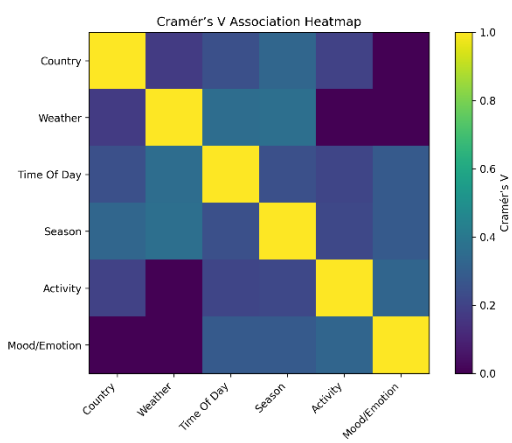


Figure 1: Dataset Cramers V Association Heatmap

Table 1: Exploratory Data Statistics Summary

Category	Metric / Value	Details
Description Length	**Average** **Minimum** **Maximum**	24.72 words 3 words 95 words
Country	**Most frequent**	Japan (36, 7.42%)
	Other countries frequency	Palestine (32, 6.60%) France (30, 6.19%) Greece (25, 5.15%) Spain (25, 5.15%) Maldives (25, 5.15%) Italy (25, 5.15%)
Weather	**Most frequent**	Sunny (276, 56.91%)
		Cloudy (100, 20.62%) Not Clear (42, 8.66%) Snowy (37, 7.63%) Clear (17, 3.51%) Rainy (10, 2.06%)
Time of Day	**Distribution**	Afternoon (243, 50.10%) Evening (133, 27.42%) Morning (109, 22.47%)
Season	**Distribution**	Summer (202, 41.65%) Spring (93, 19.18%) Winter (69, 14.23%) Fall (64, 13.20%) Not Clear (54, 11.13%)
Activity	**# Unique classes** **Most frequent** **Rare activities**	69 Sightseeing (242, 49.90%) 60 activities with <5 samples
	Examples	Sightseeing (49.90%) Relaxing (16.29%) Hiking (9.28%) No Activity (2.27%) Praying (1.65%)
Mood / Emotion	**Most frequent**	Happiness (18.97%) Curiosity (18.35%) Adventure (17.94%)
Activity × Time of Day	**Dominant trend**	Sightseeing mostly in Afternoon (50.83%) Hiking more frequent in Morning (57.78%) Relaxing mostly in Afternoon (54.43%)
Overall Challenge	**Data imbalance**	Severe class imbalance in Activity labels (long-tail distribution)

Figure 2: Visualization: Exploratory Data Analysis for activity & time of day prediction task

The exploratory data analysis revealed a significant class imbalance in activity labels, with a small number of activities dominating the dataset and many rare classes having very few samples. In contrast, the *Time-of-Day* categories were more evenly distributed. Moderate correlations were observed between activity and time of day, indicating partial dependency between the two tasks. These findings highlight key challenges such as class imbalance and data sparsity.

Experiments and Results

Before training any classifier, we used a pre-trained ResNet-50 network as a feature extractor. Each image was resized to 224×224 pixels, converted to a tensor, and normalized using ImageNet statistics. we then used a pre-trained ResNet-50 model, removing its final classification layer, to extract 2048-dimensional feature vectors from each image. These embeddings serve as rich, high-level representations of the images and were saved as a feature matrix for subsequent machine learning experiments on activity and time-of-day classification.

Baseline model: K- nearest neighbor

We implemented a k-Nearest Neighbors (k-NN) classifier using precomputed ResNet-50 embeddings for image classification, training separate models for predicting Time of Day and Activity. We also explored multi-output prediction but found similar results. For evaluation, we used cross-validation with StratifiedKFold for Time of Day and KFold for Activity due to class imbalance. In the Activity task, less frequent classes were grouped into an "other" category to simplify confusion matrix interpretation. Performance was measured using macro-averaged accuracy, precision, recall, and F1-score. Confusion matrices and sample tables of actual vs. predicted labels were generated to illustrate model behavior as shown in *Figure3* and *Table3* below.

Table 2: Model 1: KNN evaluation metrics

Task	k	Accuracy macro mean	Precision macro mean	Recall macro mean	f1 macro mean	n_classes	n_folds
Predict Time of Day	1	0.562887	0.540442	0.535526	0.537077	3	5
	3	0.546392	0.523453	0.490156	0.497036	3	5
Predict Activity	1	0.457732	0.043757	0.048051	0.043672	69	5
	3	0.486598	0.029187	0.034321	0.029491	69	5

Results and Interpretation:

- **Time of Day Prediction:** The 1-NN classifier achieved a mean accuracy of **56.3%** (5-fold CV) with macro F1 ≈ 0.54. Increasing k to 3 slightly decreased accuracy to **54.6%**, reflecting that nearest neighbors better capture the fine-grained distinctions in this small 3-class task. The confusion matrices shown in *Figure3* below show that Afternoon was the most accurately predicted class, while Morning and Evening were more frequently confused.
- **Activity Prediction:** The model struggled due to the **long-tail distribution** of activities, with 69 unique classes. The 1-NN classifier achieved a low mean accuracy of **45.8%** and macro F1 ≈ 0.044, which slightly improved to **48.7%** accuracy with k=3. Confusion matrices in *Figure3* confirm that the top activities (e.g., Sightseeing, Relaxing, Hiking) are predicted more reliably, whereas rare classes are mostly misclassified into the "other" category.

Overall, the k-NN baseline demonstrates that embeddings capture useful information for coarse-grained tasks (Time of Day) but are insufficient for high-cardinality, imbalanced tasks like Activity prediction.

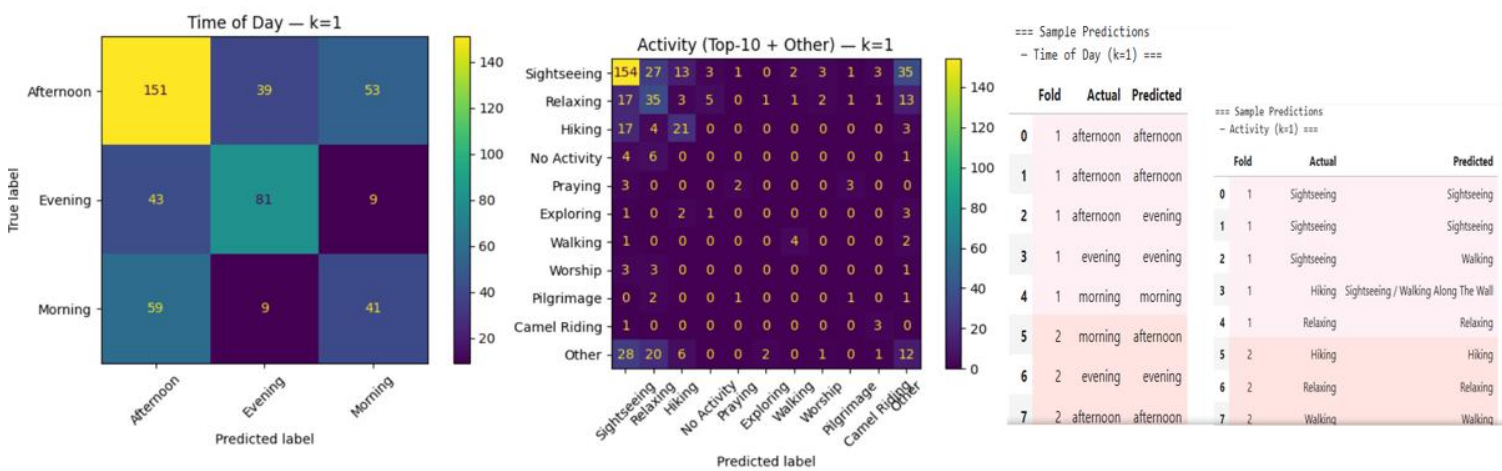
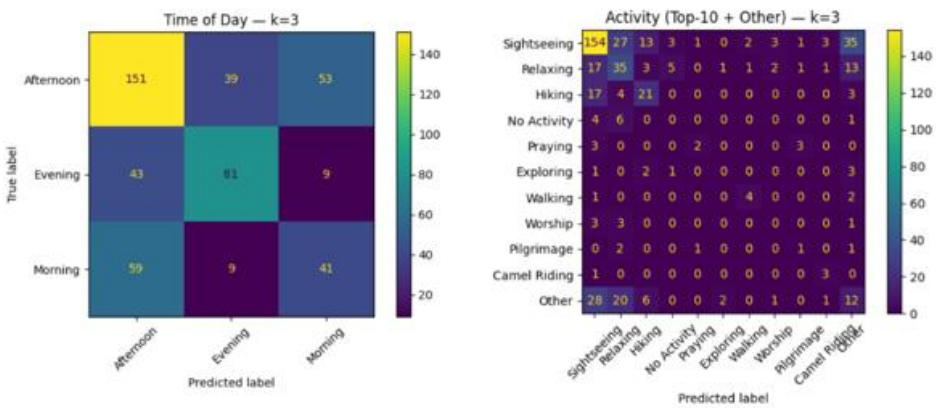


Figure 3: 1-NN Evaluation results



=== Sample Predictions
– Time of Day (k=3) ===

	Fold	Actual	Predicted
0	1	afternoon	afternoon
1	1	afternoon	afternoon
2	1	afternoon	evening
3	1	evening	evening
4	1	morning	evening
5	2	morning	afternoon
6	2	evening	evening
7	2	afternoon	morning

=== Sample Predictions
– Activity (k=3) ===

	Fold	Actual	Predicted
0	1	Sightseeing	Sightseeing
1	1	Sightseeing	Sightseeing
2	1	Sightseeing	Relaxing
3	1	Hiking	Nature Viewing
4	1	Relaxing	Walking
5	2	Hiking	Hiking
6	2	Relaxing	Relaxing

Figure 4: 3-NN Evaluation results

Model 2: Random forest

For the second model, we implemented a **Random Forest (RF) classifier** using the same ResNet-50 embeddings as input features. Random Forest is an ensemble of decision trees and naturally handles multi-class classification, making it suitable for both **Time of Day** and **Activity** prediction. To account for class imbalance, we first **collapsed rare activity classes** (those with fewer than 5 samples) into a single "other" category. Additionally, we created a **Top-K Activity mapping** (top 10 most frequent activities, others grouped as "other") for simplified visualization and reporting.

We evaluated the model using **stratified 5-fold cross-validation**, which ensures each fold maintains the same class proportions. The model was trained for a range of n_estimators ([50, 100, 200, 400, 600]) to examine the effect of ensemble size on performance. Out-of-fold (OOF) predictions were recorded in *Table4* to compute **accuracy, macro-averaged precision, recall, and F1-score**. Random samples of OOF predictions were also inspected to illustrate model behavior as shown in *Table3*.

Table 3: Random Forest: Random samples of OOF predictions

=== Random OOF Test Predictions ===

	task	n_estimators	Fold	Actual	Predicted
3881	Time of Day	600	5	evening	evening
1684	Activity	100	1	Sightseeing	Sightseeing
1044	Time of Day	100	3	evening	evening
4810	Activity	600	3	Sightseeing	Sightseeing
1538	Activity	100	3	Sightseeing	Sightseeing
1034	Time of Day	100	3	afternoon	afternoon
2107	Time of Day	200	1	evening	afternoon
4261	Time of Day	600	1	morning	afternoon
4026	Time of Day	600	1	afternoon	afternoon
4063	Time of Day	600	4	afternoon	afternoon

Results & Discussions

We performed hyperparameter tuning on the **number of estimators** (n_estimators) for the **Random Forest** model, testing values of 50, 100, 200, 400, and 600 trees

Table 4: Model 2: Random forest evaluation

=== Cross-Validated Metrics ===

	task	n_estimators	avg_accuracy	avg_precision	avg_recall	avg_f1
0	Time of Day	50	0.639175	0.673209	0.544371	0.549915
1	Activity	50	0.564948	0.299380	0.254883	0.253889
2	Time of Day	100	0.639175	0.665775	0.538157	0.540008
3	Activity	100	0.575258	0.303140	0.257747	0.256700
4	Time of Day	200	0.632990	0.662879	0.530948	0.534304
5	Activity	200	0.579381	0.323451	0.258655	0.257433
6	Time of Day	400	0.639175	0.698325	0.532146	0.536143
7	Activity	400	0.581443	0.327953	0.262591	0.262804
8	Time of Day	600	0.645361	0.711973	0.534054	0.535339
9	Activity	600	0.589691	0.333318	0.266896	0.268128

- **Time of Day** prediction achieved a mean accuracy of ~64–65%, with relatively high macro precision (~0.67–0.71) and lower recall (~0.53–0.54), reflecting the modest difficulty of this 3-class task.
- **Activity** prediction was more challenging due to the long-tail distribution of classes. Accuracy ranged from ~56% to ~59%, with macro F1-scores around 0.25–0.27. The Top-K activity evaluation slightly improved metrics but the model still struggled with rare classes.
- Increasing the number of trees (n_estimators - hyperparameter tunning) slightly improved stability and precision for both tasks but did not drastically change the overall performance.

Overall, the Random Forest model provided **better performance than the k-NN baseline**, particularly for Time of Day, due to its ability to capture non-linear feature interactions. However, Activity prediction remains challenging because of **severe class imbalance and high number of rare classes**, even after collapsing rare activities or using Top-K simplification. Random OOF test samples confirmed that the model often correctly predicted common classes while misclassifying rare ones.

Model 3: Convolutional Neural Network

This implementation involves training a multi-output Convolutional Neural Network (CNN) model for two classification tasks: predicting the *time of day* and *activity*. The model is evaluated using 5-fold cross-validation to assess its generalization performance.
Key steps include:

1. **Label Encoding**: The target labels for time of day and activity are one-hot encoded.
2. **Data Splitting**: The dataset is split into 5 folds for cross-validation.
3. **CNN Model**: The model uses Conv1D layers to extract features, followed by dense layers with dropout for regularization.
4. **Training**: The model is trained on each fold, using categorical cross-entropy loss and Adam optimizer.
5. **Evaluation**: Performance metrics like accuracy, precision, recall, and F1-score are computed for each fold and averaged to provide a comprehensive evaluation of the model's performance on both tasks.

The final output includes the average metrics for both the time of day and activity predictions, offering insights into the model's effectiveness across different data subsets.

Then we tried to improve the performance using various hyperparameter, results are shown in *Table5* below:

1. **Dropout**: Dropout is applied to reduce overfitting. The rate was varied from **0.5** to **0.7**, led to noticeable improvements in accuracy and F1 score for both tasks. Higher dropout prevented the network from becoming too reliant on specific neurons, enhancing generalization.
2. **Dense Layer Reduction**: The size of the dense layer was reduced from **256** to **128** units, which improved the time-of-day prediction accuracy, precision, and recall slightly. Reducing the dense layer size helped prevent overfitting, especially we are working with small dataset.
3. **L2 Regularization**: L2 regularization was added to the Conv1D and dense layers to penalize large weights and prevent overfitting. This improved generalization, especially for the time-of-day predictions, as seen in the increased accuracy and F1 scores.

Results & Discussion

Table 5: CNN Evaluation

Configuration	Time of Day Accuracy	Time of Day Precision	Time of Day Recall	Time of Day F1 Score	Activity Accuracy	Activity Precision	Activity Recall	Activity F1 Score
Initial Model (No Hyperparameters)	0.5891	0.5791	0.5351	0.5195	0.5772	0.5591	0.4731	0.4759
Dropout = 0.5	0.6021	0.5978	0.5536	0.5533	0.5711	0.5415	0.4975	0.5018
Dropout = 0.7	0.6165	0.5852	0.5612	0.5568	0.5897	0.5665	0.5052	0.5118
Dropout = 0.7 + Dense Layer = 128**	0.6268	0.5947	0.5412	0.5337	0.5959	0.5898	0.4835	0.4940
Dropout = 0.7 + L2 Regularization	0.6330	0.5965	0.5681	0.5367	0.5938	0.5874	0.4975	0.5132

- **Accuracy**: Accuracy improved across all configurations for both tasks. The final model with dropout at 0.7 and L2 regularization provided the highest accuracy for both time of day (0.6330) and activity (0.5938).

- **Precision**: Precision improved with dropout at 0.7 and the reduction in dense layer size, indicating better performance in predicting positive classes.
- **Recall**: Recall improved with higher dropout and L2 regularization, which reduced overfitting and allowed the model to generalize better to unseen examples.
- **F1 Score**: The F1 score, which balances precision and recall, also saw improvements in the models with higher dropout rates and L2 regularization.

In summary, the hyperparameter tuning strategies, especially higher dropout rates and L2 regularization, significantly improved the model's generalization capability and overall performance.

Analysis

The best-performing model is CNN (with Dropout = 0.7 and L2 regularization) shows strong performance but still faces challenges, especially in the Activity classification task. While the model achieves good accuracy for Time of Day (63.30%). we analyzed the model errors by comparing training and validation metrics as shown in *Figure5* below, noting that there is approximately no difference which means the model generalized good on unseen data.
Also, we noticed that the model struggles with misclassifications between similar categories, like evening vs. afternoon as shown in *Table6* below. The Activity task, with 59.38% accuracy, exhibits lower precision and recall, particularly for rare activities or those with overlapping features like "Boat Riding" vs. "Swimming") as shown in *Table6* below. Misclassifications are likely due to data imbalance and insufficient feature differentiation.

```
13/13 ━━━━━━━━━━━━━━━━━━━━ 10s 757ms/step - activity_output_accuracy: 0.6418 - activity_output_loss: 0.7780 - loss: 1.8696 - time_output_accuracy: 0.7938
- time_output_loss: 0.5498 - val_activity_output_accuracy: 0.6392 - val_activity_output_loss: 0.9115 - val_loss: 2.1931 - val_time_output_accuracy: 0.680
4 - val_time_output_loss: 0.7241
```

Figure 5: Train Vs Validation Loss

Table 6: Time of Day and Activity misclassification

```
=== Random Actual vs Predicted Samples ===
```

	Actual Time of Day	Predicted Time of Day	Actual Activity	Predicted Activity
62	afternoon	morning	Hiking	Hiking
40	morning	morning	Relaxing	Relaxing
93	afternoon	afternoon	Boat Riding	Boat Riding / Swimming / Sightseeing
18	evening	afternoon	Sightseeing	Sightseeing
81	afternoon	evening	Sightseeing	Relaxing
83	afternoon	evening	Walking / City Exploring	Boarding
64	evening	evening	Religious Pilgrimage	Worship
42	afternoon	evening	Sightseeing	No Activity
10	afternoon	afternoon	Sightseeing	Worship
0	morning	morning	Sightseeing	Hiking

Conclusion & Discussion

In this assignment, we explored the multi-task classification problem of predicting both the *Time of Day* and *Activity* based on image data. Three distinct machine learning models were implemented and evaluated: k-Nearest Neighbors (k-NN), Random Forest (RF), and Convolutional Neural Networks (CNN).

The baseline k-NN model provided reasonable results for Time-of-Day prediction but struggled significantly with Activity classification due to the class imbalance and high number of unique classes. The Random Forest model, while performing better than k-NN, still faced challenges with Activity prediction, despite hyperparameter tuning and handling class imbalance. The CNN model, with hyperparameter optimization (dropout and L2 regularization), showed the best results overall, achieving a notable improvement in both accuracy and F1 score for Time of Day and Activity prediction tasks.

Despite the improvements, the models faced challenges, particularly in the Activity classification task, where class imbalance and overlapping features led to misclassifications. The model's performance was generally strong for Time of Day, but more advanced techniques or a larger dataset may be needed for improving the Activity prediction, especially for rare or ambiguous classes.

In summary, while the CNN model performed the best, it highlighted the difficulties inherent in this multi-task classification problem, including the effects of data imbalance and the complexity of predicting multiple outputs simultaneously. Future work could focus on further tuning the models, incorporating additional data sources or applying data augmentation, or applying more advanced techniques to address misclassifications and improve overall performance.